802.11n and 802.11ac Numerical Problems

1. An 802.11n device uses 40 MHz channel, 4 spatial streams, MCS 15 (64-QAM, code rate 5/6), and long guard interval (800 ns). Compute the raw PHY data rate in Mbps. For 40 MHz: 108 data subcarriers

   Answer: For 802.11n, the data rate formula:

$$\text{Rate} = \frac{(\text{data subcarriers}) \times (\text{bits per subcarrier}) \times (\text{code rate}) \times (\text{spatial streams})}{\text{symbol duration}}$$

   For 40 MHz: 108 data subcarriers. 64-QAM: 6 bits/symbol. Code rate 5/6. Long GI symbol duration $= 3.2\,\mu\text{s} + 0.8\,\mu\text{s} = 4\,\mu\text{s}$.

$$\text{Rate} = \frac{108 \times 6 \times \frac{5}{6} \times 4}{4 \times 10^{-6}} = \frac{108 \times 5 \times 4}{4 \times 10^{-6}} = \frac{2160}{4 \times 10^{-6}} = 540 \times 10^{6} \text{ bps} = 540 \text{ Mbps}.$$

2. An 802.11n station sends a frame using MCS 7 (20 MHz, 64-QAM 5/6, 1 stream, long GI). If the frame payload is 1500 bytes, MAC header is 30 bytes, FCS is 4 bytes, and PHY preamble + PLCP header take 40 μs, compute the effective throughput (excluding ACK and DIFS). Assume no errors. For 20 MHz: 52 data subcarriers

   Answer: Raw PHY rate for MCS 7: 20 MHz $\rightarrow$ 52 data subcarriers.

T(symbol) = 4 microseconds
= 3.2 + 0.8 microseconds

$$\text{Rate} = \frac{52 \times 6 \times \frac{5}{6} \times 1}{4 \times 10^{-6}} = \frac{52 \times 5}{4 \times 10^{-6}} = \frac{260}{4 \times 10^{-6}} = 65 \text{ Mbps}.$$

   Total frame bits $= (1500 + 30 + 4) \times 8 = 12272$ bits. Transmission time $= \frac{12272}{65 \times 10^{6}} = 188.8\,\mu\text{s}$. Total time = preamble $40\,\mu\text{s} + 188.8\,\mu\text{s} = 228.8\,\mu\text{s}$. Throughput $= \frac{1500 \times 8}{228.8 \times 10^{-6}} = \frac{12000}{2.288 \times 10^{-4}} = 52.45$ Mbps.

3. In a 2×2 MIMO system (802.11n), the SNR on each spatial stream is 20 dB. The channel matrix has singular values 3.5 and 1.2. Compare the achievable sum rate (Shannon capacity) with the 802.11n maximum data rate using MCS 15 (2 streams, 40 MHz, 64-QAM 5/6).

   Answer: Shannon capacity:

$$C = B \sum_{i=1}^{2} \log_2\left(1 + \lambda_i^2 \cdot \text{SNR}_i\right).$$

   SNR linear $= 10^{20/10} = 100$.

$$C = 40 \times 10^{6} \left[\log_2(1 + 3.5^2 \times 100) + \log_2(1 + 1.2^2 \times 100)\right]$$

$$= 40 \times 10^{6} \left[\log_2(1 + 1225) + \log_2(1 + 144)\right] = 40 \times 10^{6} \left[\log_2(1226) + \log_2(145)\right]$$

MPDU aggregation (A-MPDU) is a technique in Wi-Fi where multiple MAC frames are combined and sent together as one large transmission to improve efficiency.

$$= 40 \times 10^6 [10.26 + 7.18] = 40 \times 10^6 \times 17.44 = 697.6 \text{ Mbps}.$$

802.11n max rate (40 MHz, 2 streams, MCS 15, long GI):

$$\frac{108 \times 6 \times \frac{5}{6} \times 2}{4 \times 10^{-6}} = \frac{108 \times 5 \times 2}{4 \times 10^{-6}} = \frac{1080}{4 \times 10^{-6}} = 270 \text{ Mbps}.$$

Shannon capacity (697.6 Mbps) is higher than the 270 Mbps practical limit.

4. A transmitter uses A-MPDU aggregation with 64 subframes each of size 1500 bytes. The PHY rate is 130 Mbps (MCS 14, 2 streams, 20 MHz, 64-QAM 5/6, long GI). MAC overhead per subframe is 36 bytes (including delimiter, MPDU header, FCS). Compute the frame efficiency (aggregated payload / total air time). Ignore PHY preamble.

   Answer: Total payload $= 64 \times 1500 = 96000$ bytes. Total MAC overhead $= 64 \times 36 = 2304$ bytes. Total A-MPDU size $= 96000 + 2304 = 98304$ bytes $= 786432$ bits. Transmission time $= \frac{786432}{130 \times 10^6} = 6.049 \times 10^{-3}$ s $= 6.049$ ms. Efficiency $= \frac{\text{payload bits}}{\text{total transmitted bits}} = \frac{96000 \times 8}{98304 \times 8} = \frac{96000}{98304} = 0.9766$ (97.66%).

5. In 802.11n greenfield mode, the PHY preamble is 16 μs. A station sends a 4-frame A-MPDU, each frame 1500 bytes, at MCS 7 (65 Mbps). The SIFS is 16 μs, and each Block ACK takes 44 μs. Compute the total time to send the aggregate and receive the Block ACK, and the effective throughput including the SIFS and ACK. MAC overhead per frame (30 Header +4 FCS=34 bytes)

   ## Greenfield mode is a transmission mode in IEEE 802.11n where frames are sent only for 802.11n devices, without any support for older Wi-Fi standards.

   Answer: Total payload $= 4 \times 1500 = 6000$ bytes $= 48000$ bits. MAC overhead per frame (30+4=34 bytes) $\rightarrow$ total overhead $= 4 \times 34 = 136$ bytes $= 1088$ bits. Total frame bits $= 48000 + 1088 = 49088$ bits. Transmission time $= \frac{49088}{65 \times 10^6} = 0.755$ ms $= 755$ μs. Add preamble $= 16$ μs $\rightarrow$ data air time $= 771$ μs. Total time $=$ data air time + SIFS + Block ACK $= 771 + 16 + 44 = 831$ μs. Throughput $= \frac{48000}{831 \times 10^{-6}} = 57.76$ Mbps.

6. An 802.11n link uses 2 streams, 40 MHz, MCS 15 (270 Mbps). The measured PER (packet error rate) is 0.01 for 1500-byte packets. Assuming errors are independent, what is the effective goodput after retransmissions (stop-and-wait, infinite retries)? For 40 MHz: 108 data subcarriers MAC overhead per frame (30+4=34 bytes)
   Answer: Probability of success $= 1 - 0.01 = 0.99$. Average transmissions per packet $= 1/0.99 \approx 1.0101$. Frame size $= 1500 + 30 + 4 = 1534$ bytes $= 12272$ bits. Transmission time $= \frac{12272}{270 \times 10^6} = 45.45$ μs. Goodput $= \frac{1500 \times 8}{45.45 \times 10^{-6} \times 1.0101} = \frac{12000}{45.92 \times 10^{-6}} =$ 261.4 Mbps.

7. Compare the maximum throughput of 802.11n using 40 MHz channel, 4 streams, short GI (400 ns) vs long GI (800 ns). MCS 31 (64-QAM 5/6). What is the percentage gain from short GI?

   Answer: Data subcarriers for 40 MHz = 108. Long GI: symbol duration $= 3.2 + 0.8 = 4.0$ μs.

$$R_{\text{long}} = \frac{108 \times 6 \times \frac{5}{6} \times 4}{4 \times 10^{-6}} = \frac{108 \times 5 \times 4}{4 \times 10^{-6}} = \frac{2160}{4 \times 10^{-6}} = 540 \text{ Mbps}.$$

The system must choose an MCS that works for BOTH streams

So we use the weaker SNR.

Short GI: symbol duration $= 3.2 + 0.4 = 3.6\ \mu$s.

$$R_{\text{short}} = \frac{108 \times 6 \times \frac{5}{6} \times 4}{3.6 \times 10^{-6}} = \frac{2160}{3.6 \times 10^{-6}} = 600 \text{ Mbps.}$$

Gain $= \frac{600-540}{540} = 0.1111 = 11.11\%$.

8. An 802.11n transmitter uses 20 MHz, 2 streams, MCS 14 (130 Mbps). The SNR on one stream is 22 dB and on the other 18 dB. What is the approximate MCS that can be supported if we assume the lower SNR dictates the MCS for both streams? The required SNR for 64-QAM 5/6 (MCS 15 for 2 streams) is about 26 dB, for 64-QAM 3/4 (MCS 13) is 24 dB, for 16-QAM 3/4 (MCS 9) is 18 dB. Which MCS is feasible?

   Answer: Weaker stream SNR = 18 dB. MCS 9 requires 18 dB exactly, MCS 13 requires 24 dB (too high). So MCS 9 is feasible. Rate for MCS 9 (20 MHz, 2 streams):
   $$\frac{52 \times 4 \times \frac{3}{4} \times 2}{4 \times 10^{-6}} = \frac{52 \times 3 \times 2}{4 \times 10^{-6}} = \frac{312}{4 \times 10^{-6}} = 78 \text{ Mbps.}$$

9. In a 40 MHz 802.11n link, the transmitter uses MCS 23 (2 streams, 64-QAM 5/6). However, due to interference, 10% of subcarriers are unusable. Assuming the transmitter can disable those subcarriers (tone nulling), what is the new effective data rate? (Assume 108 data subcarriers normally.)

   Answer: Normal rate for MCS 23 (40 MHz, 2 streams):
   $$\frac{108 \times 6 \times \frac{5}{6} \times 2}{4 \times 10^{-6}} = \frac{108 \times 5 \times 2}{4 \times 10^{-6}} = \frac{1080}{4 \times 10^{-6}} = 270 \text{ Mbps.}$$
   Usable subcarriers $= 108 \times 0.9 = 97.2 \approx 97$. New rate $= \frac{97\times6\times\frac{5}{6}\times2}{4\times10^{-6}} = \frac{97\times5\times2}{4\times10^{-6}} = \frac{970}{4\times10^{-6}} = 242.5$ Mbps.

10. An AP transmits a 4-frame A-MPDU using MCS 23 (270 Mbps). Each frame is 1500 bytes. The Block ACK frame is 32 bytes (including MAC). The SIFS is 16 μs. Compute the total time for the burst and ACK. Then compute the effective throughput including the preamble of 20 μs (mixed mode). PHY rate for ACK is the lowest basic rate 6 Mbps. Also compute the frame and system efficiency.

    Answer: Aggregate payload $= 4 \times 1500 = 6000$ bytes $= 48000$ bits. MAC overhead per frame: 30+4=34 bytes $\rightarrow$ total overhead = 136 bytes = 1088 bits. Total A-MPDU bits = 49088 bits. Data transmission time $= \frac{49088}{270\times10^{6}} = 181.81\ \mu$s. Add preamble $= 20\ \mu$s $\rightarrow$ data burst $= 201.81\ \mu$s. SIFS $= 16\ \mu$s. Block ACK: 32 bytes = 256 bits, rate 6 Mbps $\rightarrow$ time $= \frac{256}{6\times10^{6}} = 42.67\ \mu$s. Plus preamble for ACK (20 μs) $\rightarrow$ ACK air time $= 62.67\ \mu$s. Total $= 201.81 + 16 + 62.67 = 280.48\ \mu$s. Throughput $= \frac{48000}{280.48\times10^{-6}} = 171.1$ Mbps. frame Efficiency $= \frac{48000}{49088} = 97.78\%$
    system efficiency $= \frac{171.1 \text{ Mbps}}{270 \text{ Mbps}}$ $\approx$ **63.37%**

11. Given that 802.11n supports up to 4 spatial streams, compute the increase in data rate when going from 2 streams to 4 streams for 40 MHz, MCS 15 (64-QAM 5/6), long GI. Express as factor and percentage.

    Answer: Rate for N streams $= \frac{108\times6\times5/6\times N}{4\times10^{-6}} = \frac{108\times5\times N}{4\times10^{-6}} = 135 \times N$ Mbps. For $N = 2$: 270 Mbps; $N = 4$: 540 Mbps. Increase factor $= 540/270 = 2$. Percentage increase = 100%.

12. In 802.11n, the maximum A-MPDU size is 65535 bytes. If each MPDU has a maximum of 1500 bytes payload, what is the maximum number of MPDUs that can be aggregated? Also compute the efficiency if all MPDUs are full size, with each MPDU having 36 bytes overhead (delimiter+header+FCS). Compare to the case where the last MPDU is only 500 bytes.

    Answer: Each MPDU total size = 1500+36 = 1536 bytes. Maximum number = $\lfloor 65535/1536 \rfloor = 42$ (since $42 \times 1536 = 64512$ bytes). Efficiency for full = $\frac{42 \times 1500}{64512} = \frac{63000}{64512} = 0.9766$ (97.66%). With last MPDU 500 bytes: total payload = $41 \times 1500 + 500 = 61500 + 500 = 62000$ bytes. Total aggregated size = $41 \times 1536 + (500 + 36) = 62976 + 536 = 63512$ bytes. Efficiency = $\frac{62000}{63512} = 0.9762$ (97.62%).

13. A 802.11n transmitter uses short GI (400 ns) and 20 MHz channel, 4 streams, MCS 31 (64-QAM 5/6). Compute the raw data rate. Then, if the transmitter switches to long GI but increases channel width to 40 MHz, which configuration gives higher rate?

    Answer: Short GI, 20 MHz, 4 streams: Data subcarriers = 52. Symbol time = 3.6 μs.

$$R = \frac{52 \times 6 \times \frac{5}{6} \times 4}{3.6 \times 10^{-6}} = \frac{52 \times 5 \times 4}{3.6 \times 10^{-6}} = \frac{1040}{3.6 \times 10^{-6}} = 288.89 \text{ Mbps}.$$

    Long GI, 40 MHz, 4 streams: Data subcarriers = 108, symbol time = 4 μs.

$$R = \frac{108 \times 6 \times \frac{5}{6} \times 4}{4 \times 10^{-6}} = \frac{108 \times 5 \times 4}{4 \times 10^{-6}} = \frac{2160}{4 \times 10^{-6}} = 540 \text{ Mbps}.$$

    40 MHz long GI gives higher rate (540 > 288.89 Mbps).

14. An 802.11n AP uses beamforming with 4 antennas to a single station with 2 antennas. The beamforming gain is 6 dB. The original SNR without beamforming on each stream was 20 dB. After beamforming, what is the effective SNR and the maximum possible MCS if the required SNR for 2-stream MCS 15 is 28 dB and MCS 13 is 24 dB? Assume the gain applies equally to both streams.

    Answer: New SNR = 20 + 6 = 26 dB. This is above 24 dB but below 28 dB, so MCS 13 (64-QAM 3/4) is achievable. Maximum rate: 40 MHz, 2 streams, MCS 13:

$$\frac{108 \times 6 \times \frac{3}{4} \times 2}{4 \times 10^{-6}} = \frac{108 \times 4.5 \times 2}{4 \times 10^{-6}} = \frac{972}{4 \times 10^{-6}} = 243 \text{ Mbps}.$$

15. Compute the raw PHY rate for 802.11ac using 80 MHz channel, 3 spatial streams, MCS 9 (256-QAM, code rate 5/6), short GI (400 ns). Number of data subcarriers for 80 MHz is 234.

    Answer: Data subcarriers = 234. 256-QAM $\rightarrow$ 8 bits/symbol. Code rate = 5/6. Streams = 3. Short GI symbol = 3.2 + 0.4 = 3.6 μs.

$$\text{Rate} = \frac{234 \times 8 \times \frac{5}{6} \times 3}{3.6 \times 10^{-6}} = \frac{234 \times \frac{40}{6} \times 3}{3.6 \times 10^{-6}} = \frac{234 \times 20}{3.6 \times 10^{-6}} = \frac{4680}{3.6 \times 10^{-6}} = 1.3 \times 10^9 \text{ bps} = 1300 \text{ Mbps}$$

16. In 802.11ac, the VHT preamble has a duration of 40 μs for 80 MHz (including L-STF, L-LTF, VHT-SIG-A, VHT-STF, VHT-LTF, VHT-SIG-B). A station sends a single MPDU of 1500 bytes at MCS 5 (64-QAM 2/3, 1 stream, 80 MHz, long GI). Compute the total air time and the effective throughput including preamble. Number of data subcarriers for 80 MHz is 234. MAC overhead per frame (30+4=34 bytes)

    Answer: MCS 5: 80 MHz, 1 stream, 64-QAM (6 bits), code rate 2/3. Data subcarriers = 234. Long GI symbol = 4 μs.

$$\text{Rate} = \frac{234 \times 6 \times \frac{2}{3} \times 1}{4 \times 10^{-6}} = \frac{234 \times 4}{4 \times 10^{-6}} = \frac{936}{4 \times 10^{-6}} = 234 \text{ Mbps.}$$

    Frame bits $= (1500 + 30 + 4) \times 8 = 1534 \times 8 = 12272$ bits. Data transmission time $= \frac{12272}{234\times 10^6} = 52.44\ \mu$s. Add preamble 40 μs $\rightarrow$ total $= 92.44\ \mu$s. Throughput $= \frac{1500\times 8}{92.44\times 10^{-6}} = \frac{12000}{9.244\times 10^{-5}} = 129.8$ Mbps.

17. An 802.11ac AP uses MU-MIMO to transmit to three stations simultaneously. The AP has 4 antennas. Station A supports 2 streams, Station B 1 stream, Station C 1 stream. The AP uses 80 MHz channel, MCS 7 (64-QAM 5/6) for all streams. Compute the total PHY rate and the per-station rates. Assume long GI. Number of data subcarriers for 80 MHz is 234.

    Answer: Total streams $= 2+1+1 = 4$. For 80 MHz, MCS 7 per-stream rate (long GI):

$$\frac{234 \times 6 \times \frac{5}{6}}{4 \times 10^{-6}} = \frac{234 \times 5}{4 \times 10^{-6}} = \frac{1170}{4 \times 10^{-6}} = 292.5 \text{ Mbps.}$$

    Station A (2 streams): $2 \times 292.5 = 585$ Mbps. Station B: 292.5 Mbps. Station C: 292.5 Mbps. Total $= 585 + 292.5 + 292.5 = 1170$ Mbps.

18. Compare the maximum throughput of 802.11ac using 80 MHz vs 160 MHz, 8 streams, MCS 9 (256-QAM 5/6), short GI. Also compute the percentage gain. Data subcarriers: 80 MHz $\rightarrow$ 234, 160 MHz $\rightarrow$ 468.

    Answer: Data subcarriers: 80 MHz $\rightarrow$ 234, 160 MHz $\rightarrow$ 468. Per-stream rate for 80 MHz short GI:

$$\frac{234 \times 8 \times \frac{5}{6}}{3.6 \times 10^{-6}} = \frac{234 \times 6.6667}{3.6 \times 10^{-6}} = \frac{1560}{3.6 \times 10^{-6}} = 433.33 \text{ Mbps.}$$

    8 streams: $433.33 \times 8 = 3466.67$ Mbps (3.467 Gbps). For 160 MHz:

$$\frac{468 \times 8 \times \frac{5}{6}}{3.6 \times 10^{-6}} = \frac{468 \times 6.6667}{3.6 \times 10^{-6}} = \frac{3120}{3.6 \times 10^{-6}} = 866.67 \text{ Mbps/stream.}$$

    8 streams: $866.67 \times 8 = 6933.33$ Mbps (6.933 Gbps). Gain $= \frac{6933.33-3466.67}{3466.67} = 1.0 =$ 100%.

19. In 802.11ac, the VHT-SIG-B field length depends on the number of streams. For 4 streams, the VHT-SIG-B takes 4 μs. The VHT-LTF duration is 4 μs per stream (for 4 streams, $4 \times 4 = 16\ \mu$s). The other preambles (L-STF, L-LTF, VHT-SIG-A, VHT-STF) sum to 28 μs. Compute total preamble duration for a 4-stream

transmission. Then compute the efficiency (payload time / total time) for a 1500 byte packet at 1 Gbps PHY rate.

Answer: Total preamble = 28 μs (other) + VHT-LTF (16 μs) + VHT-SIG-B (4 μs) = 48 μs. Payload + MAC overhead = 1534 bytes = 12272 bits. At 1 Gbps, transmission time = 12.272 μs. Total time = 48 + 12.272 = 60.272 μs. Efficiency = payload time / total time = $\frac{12.272}{60.272}$ = 0.2036 (20.36%).

20. An 802.11ac AP uses 80 MHz channel and serves 4 clients with 1 stream each using MU-MIMO (total 4 streams). Each client uses MCS 8 (256-QAM 3/4) short GI. Compute total PHY rate. If one client experiences interference and falls back to MCS 4 (16-QAM 1/2), while others remain at MCS 8, what is the new total rate? (Assume independent per-user MCS.)

    Answer: Per-stream rate for MCS 8, 80 MHz short GI:

    $$\frac{234 \times 8 \times \frac{3}{4}}{3.6 \times 10^{-6}} = \frac{234 \times 6}{3.6 \times 10^{-6}} = \frac{1404}{3.6 \times 10^{-6}} = 390 \text{ Mbps}.$$

    Total with 4 streams at MCS8 = 4 × 390 = 1560 Mbps. MCS 4 per-stream rate (16-QAM 1/2):

    $$\frac{234 \times 4 \times \frac{1}{2}}{3.6 \times 10^{-6}} = \frac{234 \times 2}{3.6 \times 10^{-6}} = \frac{468}{3.6 \times 10^{-6}} = 130 \text{ Mbps}.$$

    New total = $3 \times 390 + 130 = 1170 + 130 = 1300$ Mbps.